Warmup: Similar triangles

One fun trick that we often use to relate variables in related rates problems is a thing you probably learned in high school but maybe forgot about: *similar triangles*. This warmup will help you see what this is and why it works.

1. Draw a big triangle, any shape you want. Could be equilateral, could be right, could be scalene. Label its sides A, B, and C, and measure each side with a ruler.

2. Inside the big triangle, draw a line parallel to one of the sides. You have thus made a smaller triangle inside the big triangle. Label the sides of this triangle a, b, and c, corresponding to your labeling of the big triangle, and measure each side with a ruler.

3. Compute a bunch of ratios, then make some observations:

$$\frac{A}{a} = \qquad \frac{B}{b} = \qquad \frac{C}{c} =$$

$$\frac{A}{B} = \qquad \frac{A}{C} = \qquad \frac{B}{C} =$$

$$\frac{a}{b} = \qquad \frac{a}{c} = \qquad \frac{b}{c} =$$